Queue Implementation via Circular Array

	0	1	2	3
arr	50	20	30	40
		f, r		

q.add(10)
q.add(20)
q.add(30)
q.remove()
q.add(40)
q.add(50)

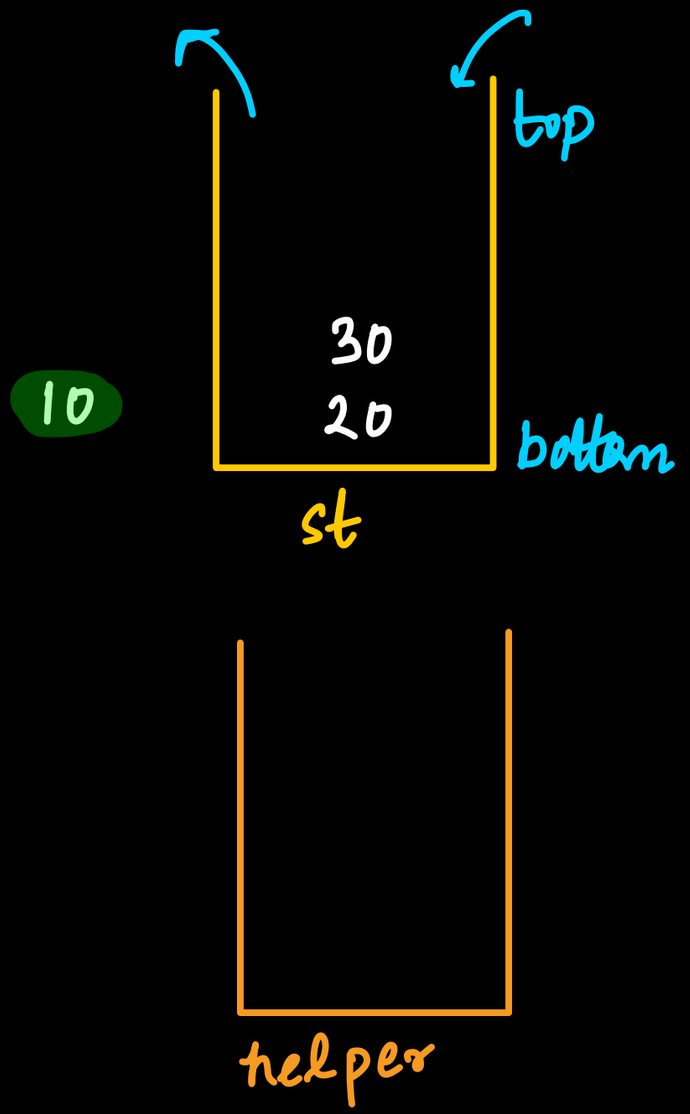
remove → f++
add → arr[r++] = val

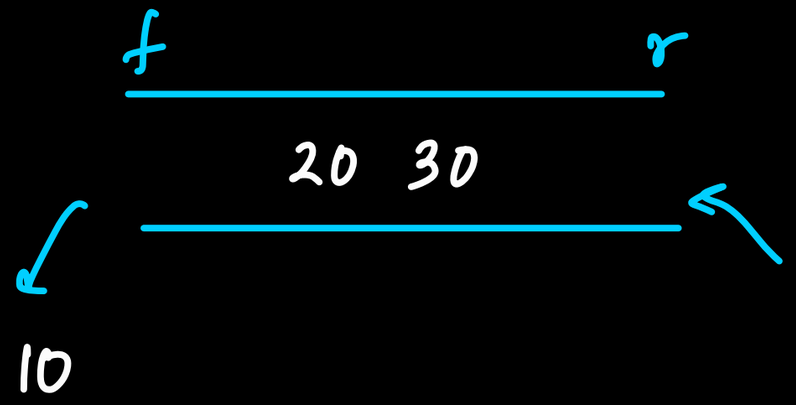
f _____ r

20 30 40 50

Ques: Implement Queue using Stacks

add efficient approach





O(1) q.add(val) → st.push(val)

O(n) q.remove() → st ka bottom remove karna hai

& q.peek() ↗

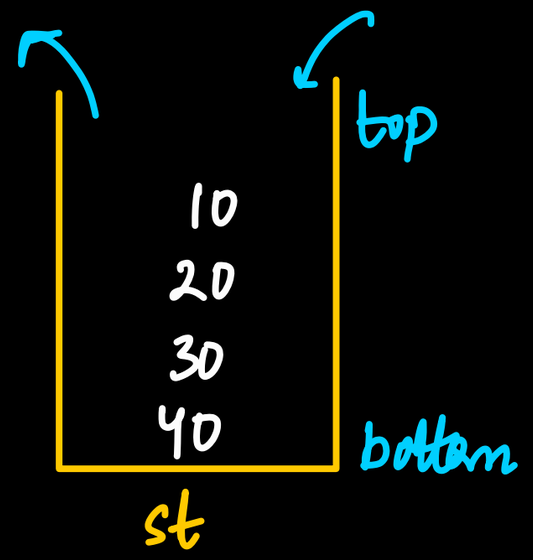
O(n) space

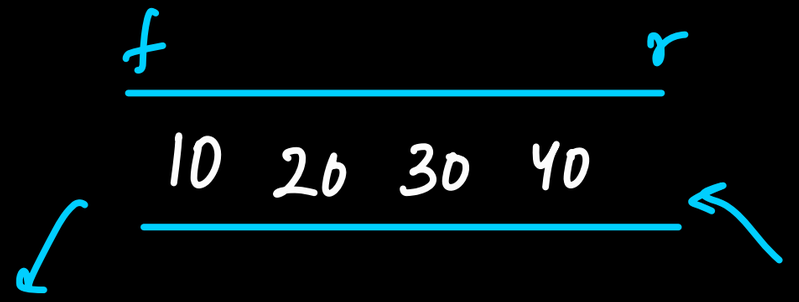
Leetcode 232

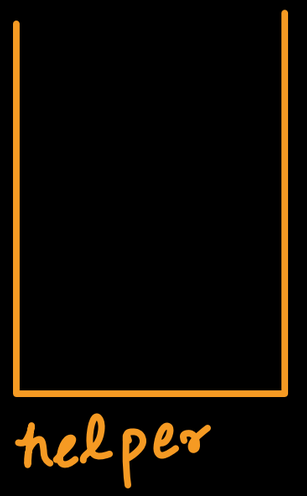
FIFO

Ques: Implement Queue using Stacks

pop & peek efficient approach







q.remove → st.pop()

q.add → st push at bottom

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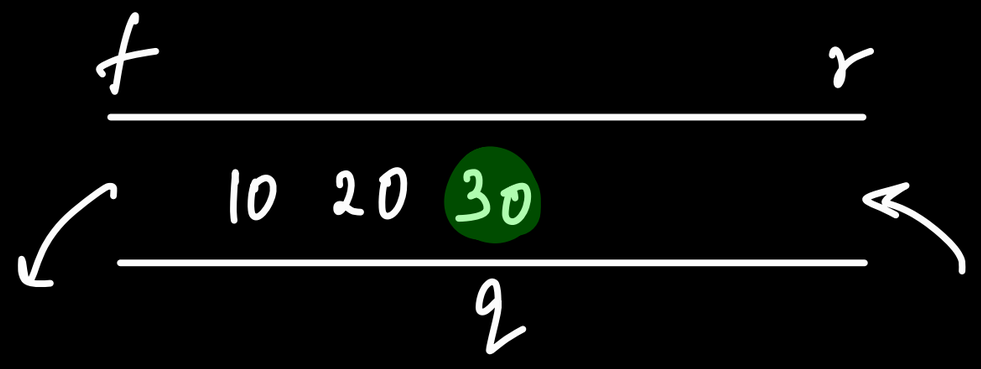
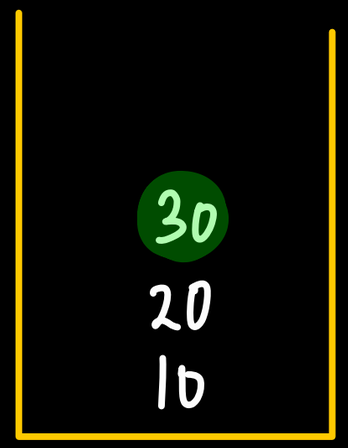
Ques: Implement Stack using Queues

add efficient approach



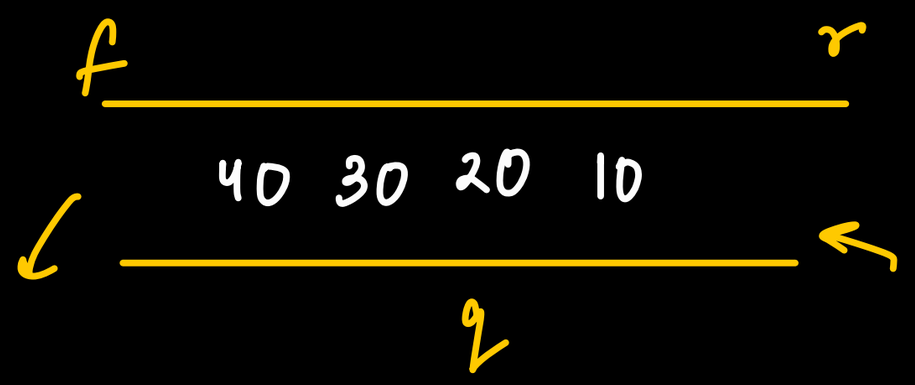
st.pop() → q ka rear ele remove karna

st.push() → q.add



Leetcode 225

Ques: Implement Stack using Queues

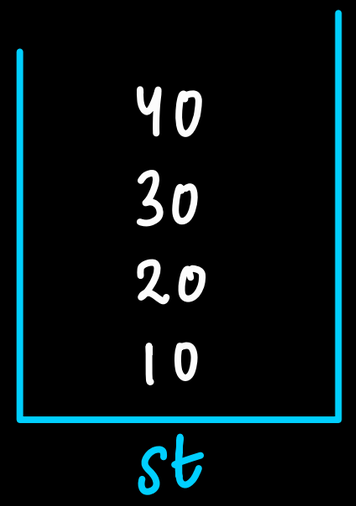


pop efficient approach
& peek

st.push → q ke front pe daalo

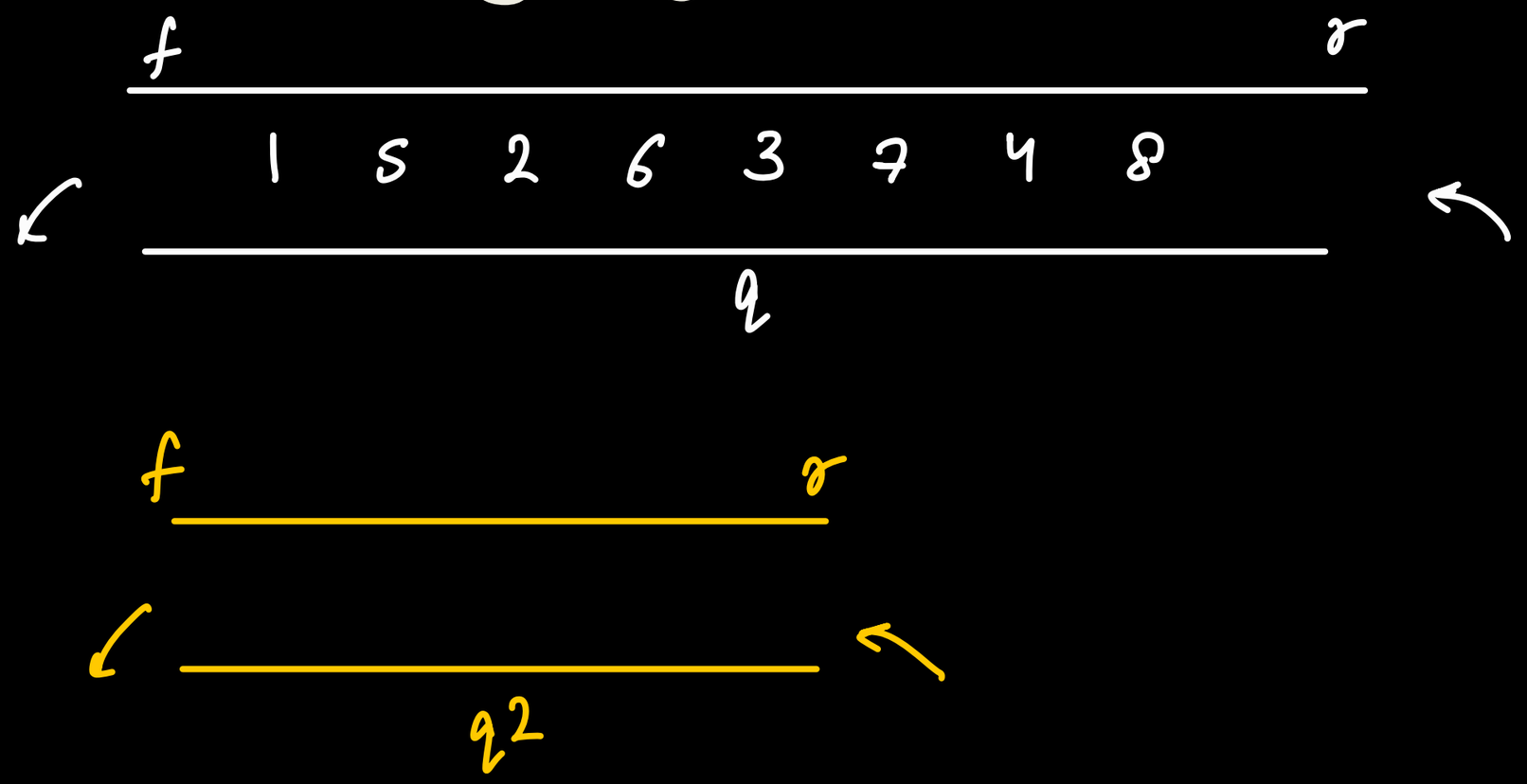
st.pop → q.remove

st.peek → q.peek



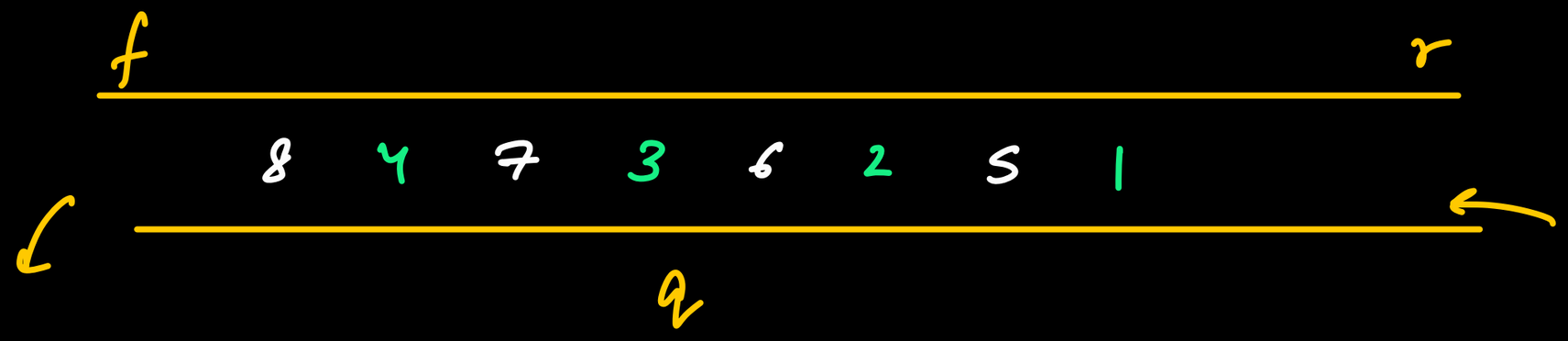
Leetcode 225

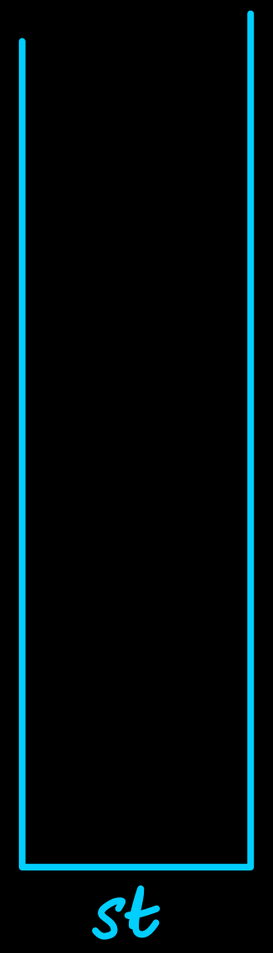
Ques: Rearrange Queue



GFG Practice

Ques: Rearrange Queue





Steps :

Put n/2 elements from q to st

Put from st to q

Put n/2 elements from q to st

** st se q me & q se q me

Reverse the q

GFG Practice